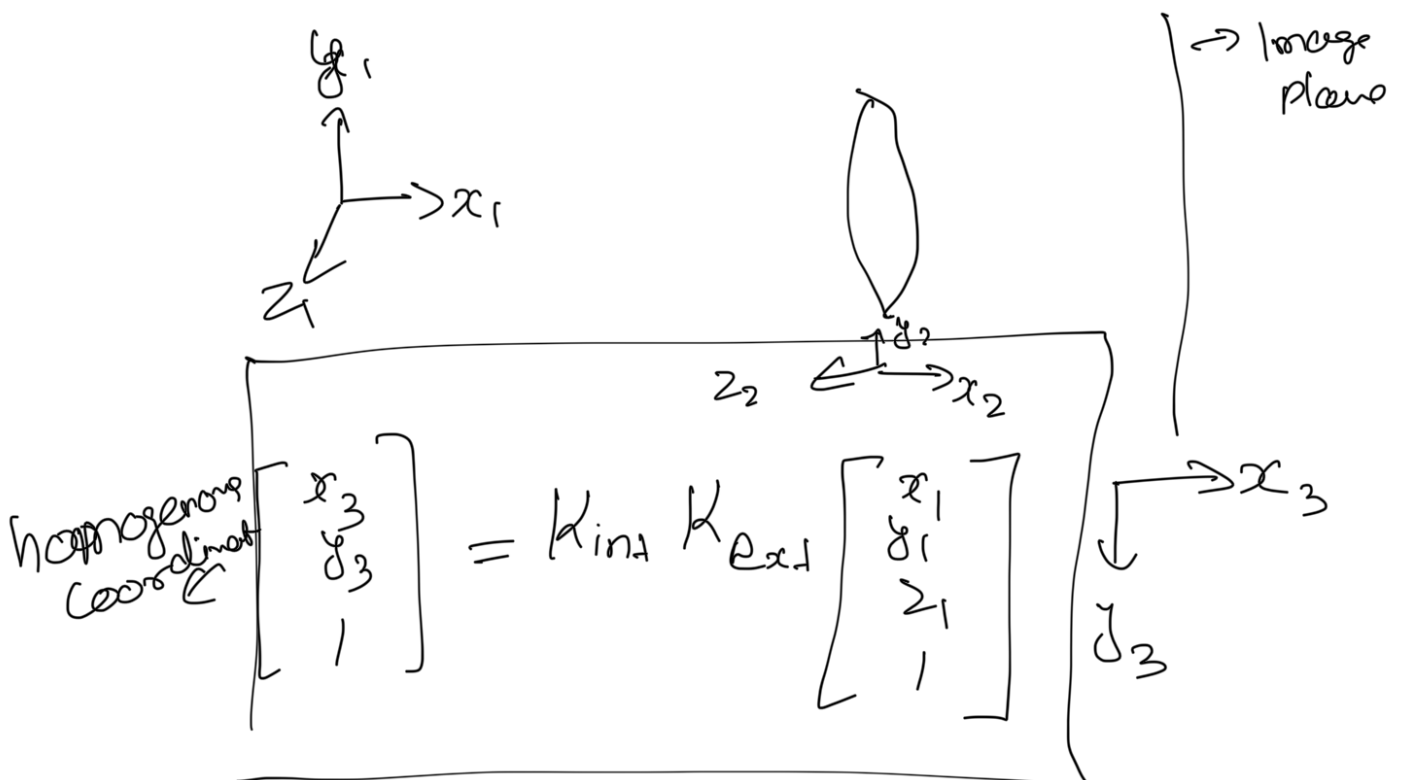
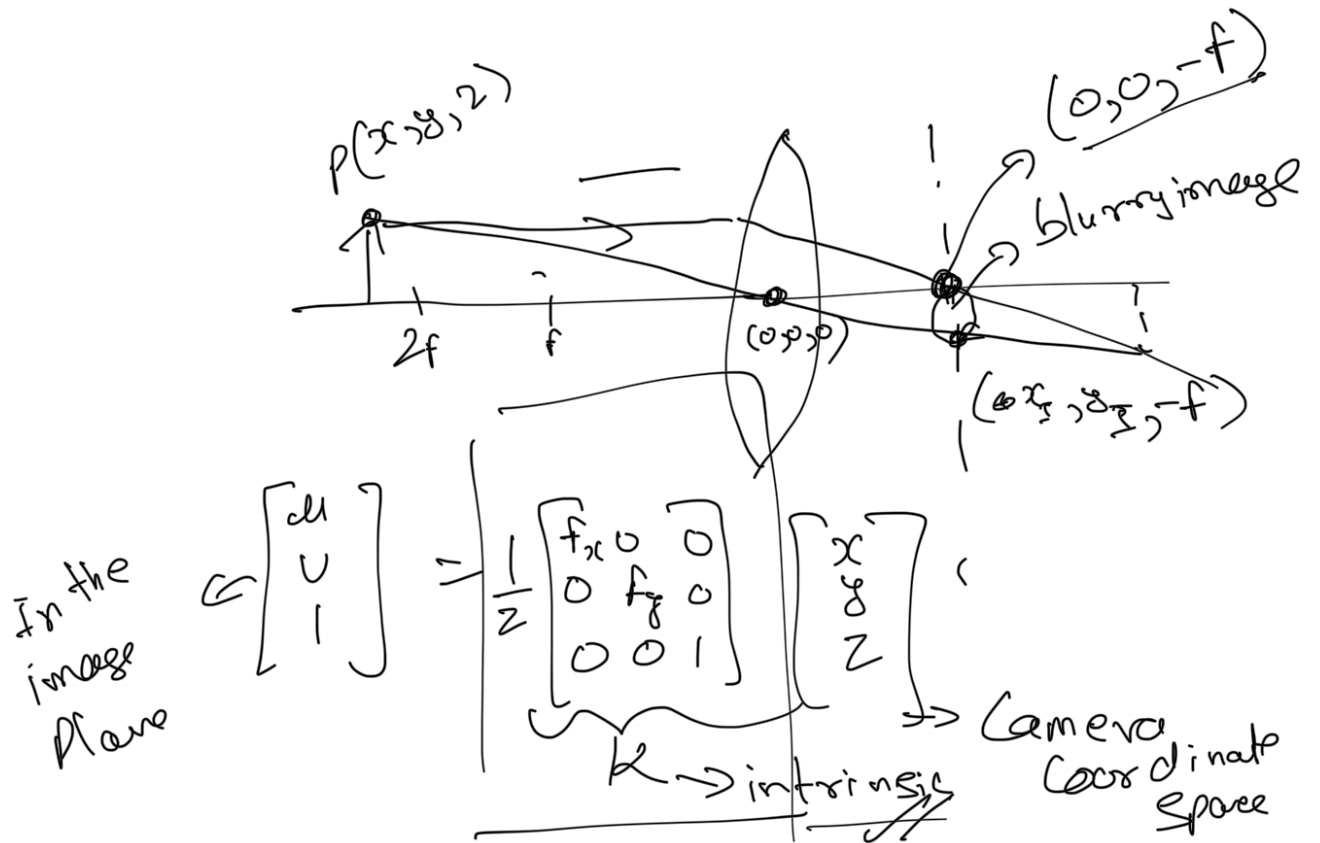
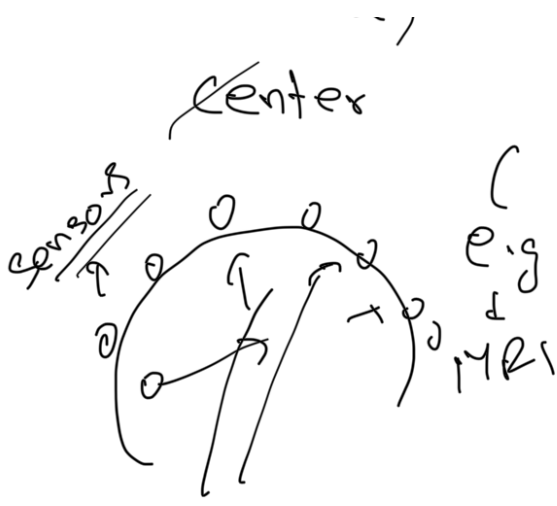




Don't  $u, v$  are in mm; we want in pixel  
 what it  $\Rightarrow$  image plane



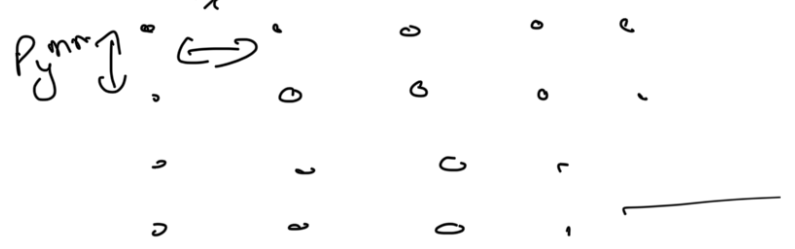
is not at  $(0,0,-f)$   
 e.g.  
~~that is not at  $(0,0,-f)$~~   
 that image plane is at  $(-c_x, -c_y, -f)$  plane center is at  $(0,0,-f)$

mm  
mm  
Pixels

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \frac{1}{z} \begin{bmatrix} f_x & 0 & 0 \\ 0 & f_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Coordinates in ~~the~~ Cam or coordinate

Assume  $\Rightarrow$  Pixel grid is as follows system



$P_x$   
 $P_y$

$$x_p = \frac{u}{P_x}$$

$$y_p = \frac{v}{P_y}$$

$\Rightarrow$  i.e. scale your coordinate in x direction with  $1/P_x$

in y direction

$$\begin{bmatrix} x_p \\ y_p \end{bmatrix} = \begin{bmatrix} f_x & 0 & 0 \\ 0 & f_y & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

with  $1/P_y$

$$\underbrace{\begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix}}_{\text{Pixel Units}} = \frac{1}{z} \underbrace{\begin{bmatrix} \frac{f}{p_x} & 0 & 0 \\ 0 & \frac{f}{p_y} & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{K_{int}} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\text{Camera Coordinates}}$$

a) Image plane center is not at  $(0, 0, -f)$  but is at

$$\underbrace{(-c_x, -c_y, -f)}$$

In image plane, translate focus center point by  $(c_x, c_y)$

In matrix form

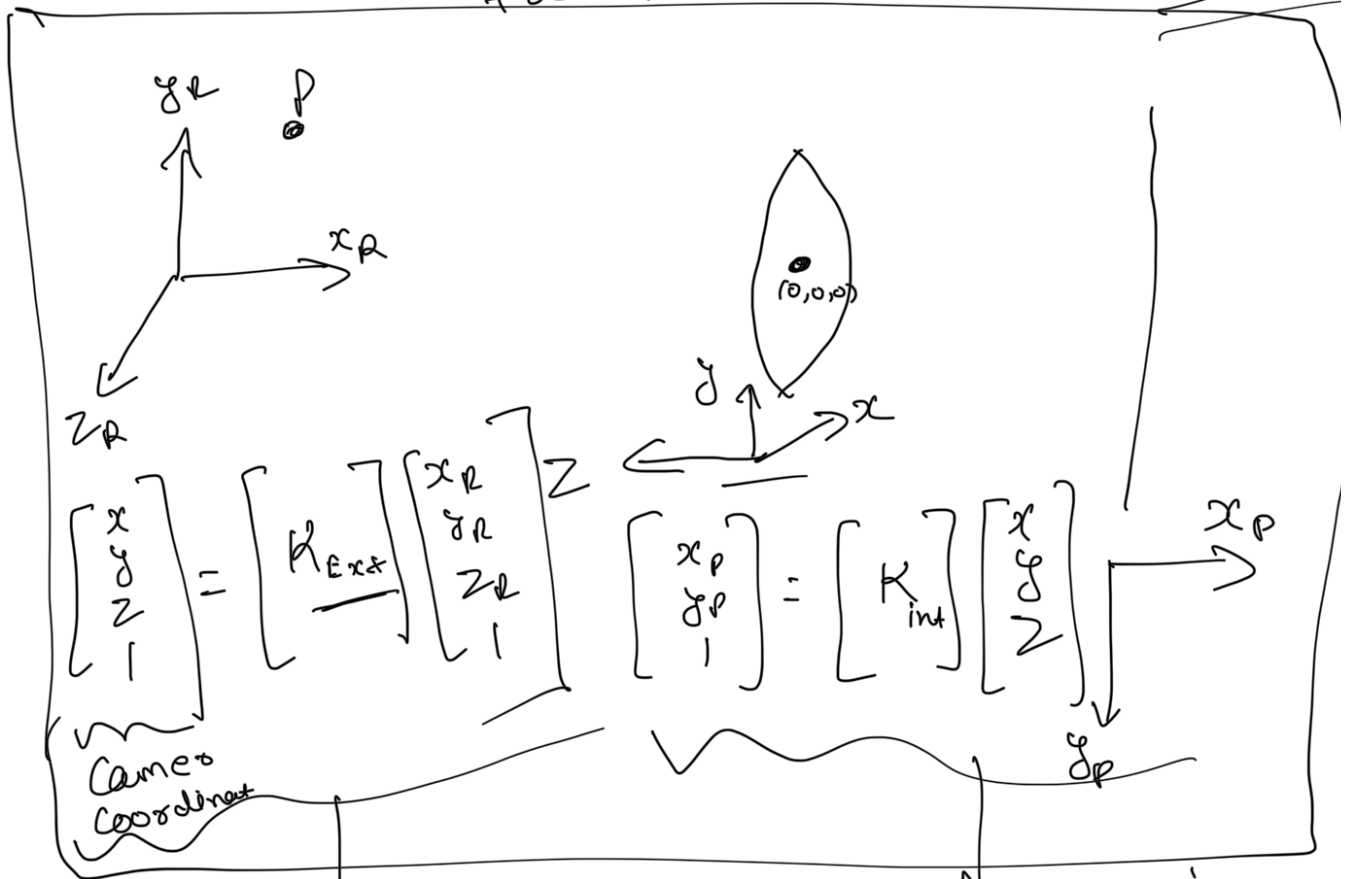
$$\underbrace{\begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix}}_{\text{In image plane}} = \frac{1}{z} \underbrace{\begin{bmatrix} \frac{f}{p_x} & 0 & c_x \\ 0 & \frac{f}{p_y} & c_y \\ 0 & 0 & 1 \end{bmatrix}}_{\substack{K_{int} \\ \text{Intrinsic matrix for camera calibration}}} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\text{Camera Coordinates}}$$

$$x_p = \frac{f}{p_x} \times \frac{1}{z} x + c_x$$

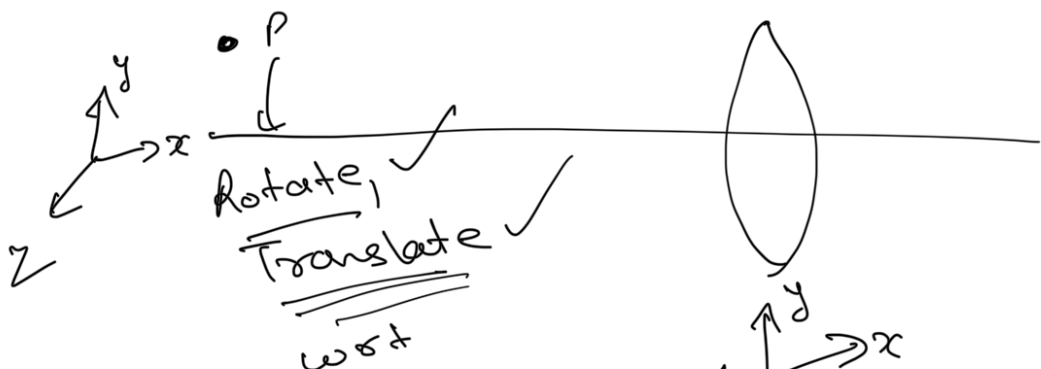
$$y_p = \frac{f}{p_y} \times \frac{1}{z} y + c_y$$

Perspective transformation

} => "Heart of Computer Graphics"



Extrinsic matrix



To Camera Coordinate  $\leftarrow$

1) Rotation

Rotation in XY plane  $\Rightarrow$  about z-axis

$$R_z = \begin{bmatrix} \cos \theta_z & -\sin \theta_z & 0 \\ \sin \theta_z & \cos \theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation in YZ plane  $\Rightarrow$  about x-axis

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_x & -\sin \theta_x \\ 0 & \sin \theta_x & \cos \theta_x \end{bmatrix}$$

Rotation in ZX plane  $\Rightarrow$  about y-axis

$$R_y = \begin{bmatrix} \cos \theta_y & 0 & \sin \theta_y \\ 0 & 1 & 0 \\ -\sin \theta_y & 0 & \cos \theta_y \end{bmatrix}$$

Orthogonal matrix  $\underbrace{A^T A = I}$

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}_{3 \times 3} = \underline{R_z R_x R_y}$$

$t_x$   
 $t_y$   
 $t_z$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} = \begin{bmatrix} x_R \\ y_R \\ z_R \end{bmatrix}$$

$$t = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}_{3 \times 1}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3 \times 1} = \begin{bmatrix} R & t \end{bmatrix}_{3 \times 4} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}_{4 \times 1}$$

Camera coordinate      Extrinsic Matrix

Camera Calibration

↓ finding  $K_{int}$  and  $K_{ext}$

$$\begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix}_{3 \times 1} = \underbrace{K_{int}}_{3 \times 3} \underbrace{K_{ext}}_{3 \times 4} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}_{4 \times 1}$$

Image coordinate      Real world coordinate

Imaging is not lossy in ideal conditions

distortion

- Lenses are perfect
- no aberrations
- no effect of ambient light
- Image formed exactly at focus
- no diffraction effect
- ~~at~~ refractive index (uniform)

/

of lens is uniform